Module 1 CNS QB Solved

1. Discuss the fundamental characteristics of data communication. With a neat diagram, explain the components of a network communication system. (8 Marks)

Answer

A. Fundamental Characteristics of Data Communication

Data communication is the exchange of data between two devices through some form of transmission medium. For effective data communication, the following four fundamental characteristics must be satisfied:

1. Delivery

- Data must be delivered to the correct destination.
- The data should reach the intended receiver and not any other device.

2. Accuracy

- Data must be delivered accurately without any alteration during transmission.
- Errors in the transmitted data should be avoided.

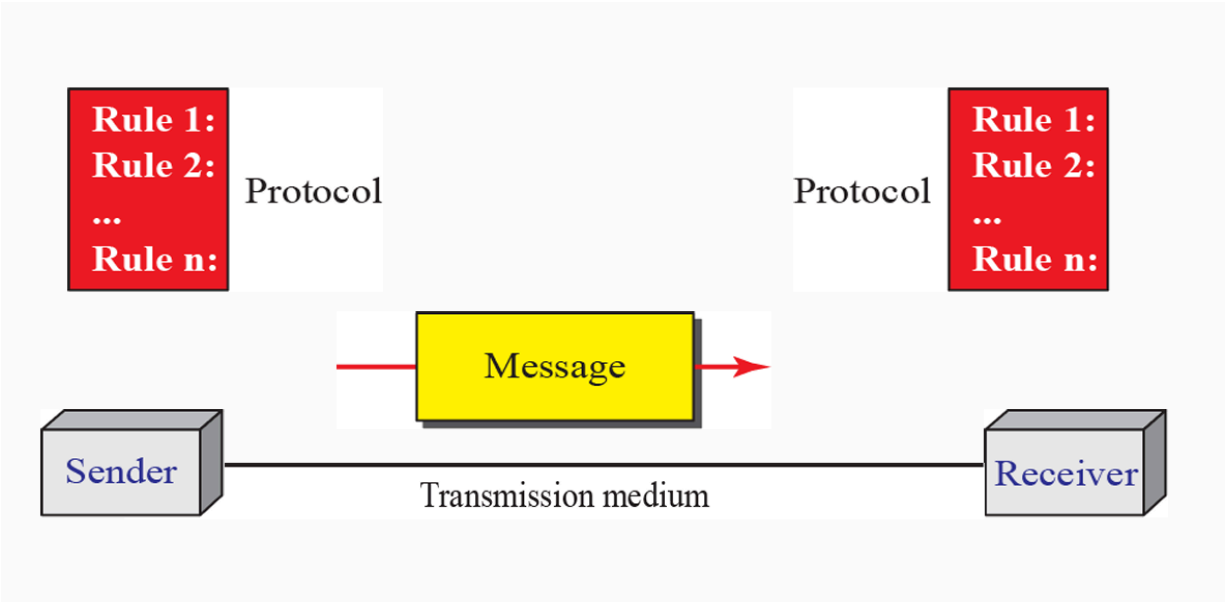
3. Timeliness

- Data must be delivered within the required time.
- Delayed data may become useless, especially in real-time applications such as audio and video communication.

4. Jitter

- Jitter refers to the variation in packet arrival time.
- It should be minimized in real-time applications to ensure smooth audio and video transmission.

B. Components of a Network Communication System



A data communication system consists of the following five components:

1. Message

- The message is the data or information to be communicated between devices.
- It may be in the form of text, numbers, images, audio, or video.

2. Sender

- The sender is the device that sends the data message.
- It may be a computer, mobile phone, or any other communication device.

3. Receiver

- The receiver is the device that receives the message sent by the sender.
- It processes the received data and makes it available to the intended user.

4. Transmission Medium

- The transmission medium is the physical path through which the message is transmitted from the sender to the receiver.
- Examples include:
    - Twisted-pair cable
    - Coaxial cable
    - Radio waves
    - Fiber-optic cable

5. Protocol

- A protocol is a set of rules that governs data communication.
- It represents an agreement between communicating devices.
- It defines how data is transmitted and received.
- Without a protocol, two devices may be connected but cannot communicate effectively.

2. Discuss the different modes of communication between two devices with suitable diagrams and examples. (8 Marks)

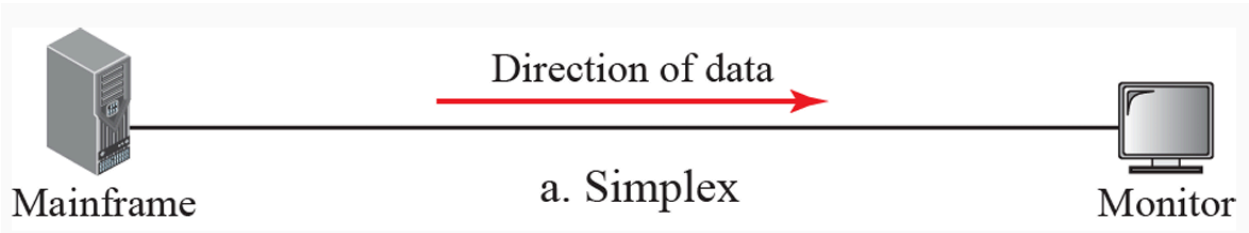
Answer

Data communication between two devices can take place in three different modes depending on the direction of data flow. They are:

- Simplex
- Half-duplex
- Full-duplex

1. Simplex Mode

In simplex mode, communication takes place in only one direction. One device acts as the sender, and the other device acts as the receiver. The receiver cannot send data back to the sender.



Characteristics:

- Data flows in only one direction.
- The entire channel capacity is used for one-way communication.
- The receiver cannot transmit data to the sender.

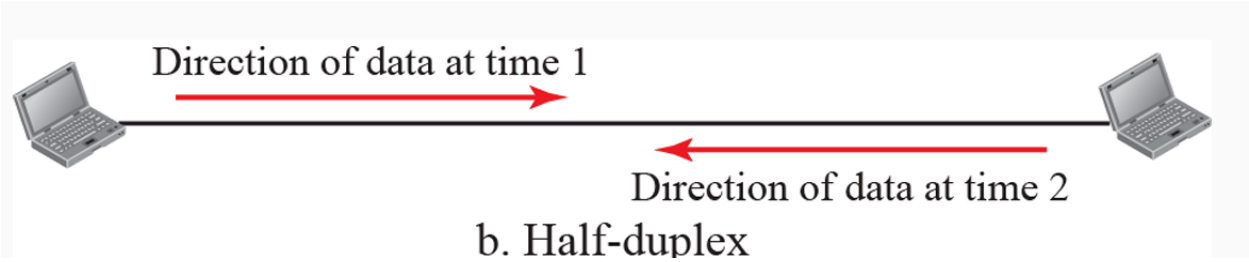
Examples:

- Keyboard to computer.

- Television broadcasting.
- Radio broadcasting.

2. Half-Duplex Mode

In half-duplex mode, communication takes place in both directions, but not at the same time. Each device can transmit and receive data, but only one device can transmit at a time.



b. Half-duplex

Characteristics:

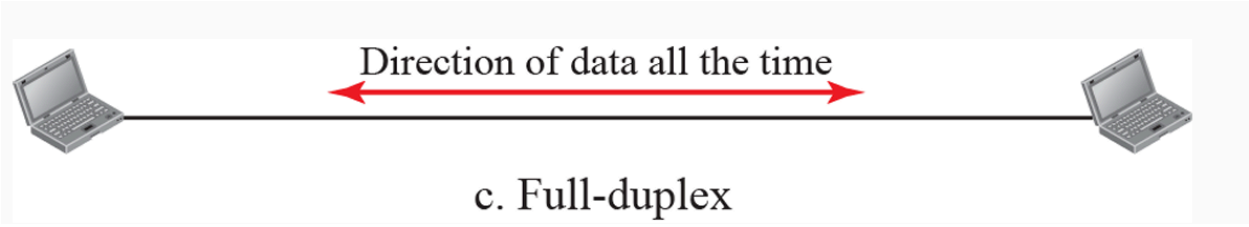
- Data flows in both directions.
- Both devices can transmit and receive data.
- Communication cannot take place simultaneously.
- The channel capacity is shared between the two directions.

Examples:

- Walkie-talkies.
- CB (Citizens Band) radio communication.

3. Full-Duplex Mode

In full-duplex mode, communication takes place in both directions simultaneously. Both devices can transmit and receive data at the same time.



c. Full-duplex

Characteristics:

- Data flows in both directions simultaneously.

- Both devices can transmit and receive data at the same time.
- It provides faster and more efficient communication.
- The channel capacity is divided between the two directions or separate paths are used.

Examples:

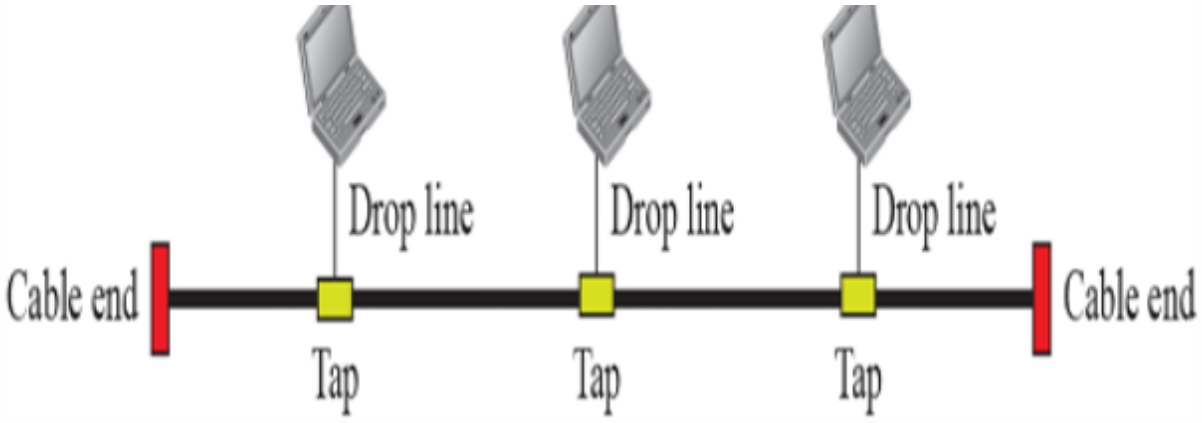
- Telephone conversation.
- Video calling.
- Modern switched Ethernet.

I'll give you the complete answer in a clean, copy-ready format. The question, key technical terms, and subheadings will all be bolded using actual Markdown bold—not just visual styling.

3. Explain different types of Topologies in communication networks with Pros and cons.

Answer:

1. Bus Topology



In Bus Topology, all devices are connected to a single central cable called the backbone. Data transmitted by a device travels along the backbone and is received by the intended device. Terminators are placed at both ends of the cable to prevent signal reflection.

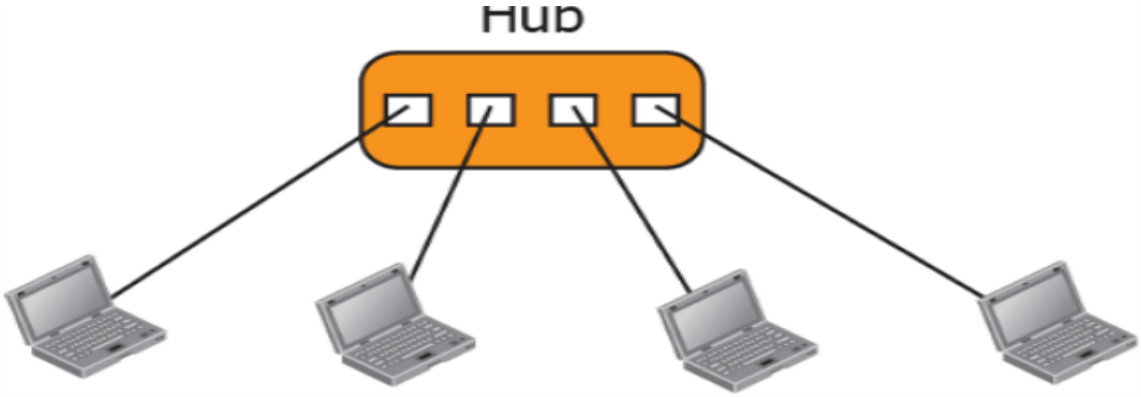
Pros:

- Simple and easy to install.
- Requires less cable than many other topologies.
- Cost-effective for small networks.

Cons:

- Failure of the backbone cable brings down the entire network.
- Difficult to identify and troubleshoot faults.
- Performance decreases as the number of devices increases.

2. Star Topology



In Star Topology, all devices are connected to a central device such as a hub or switch. All communication between devices passes through the central device.

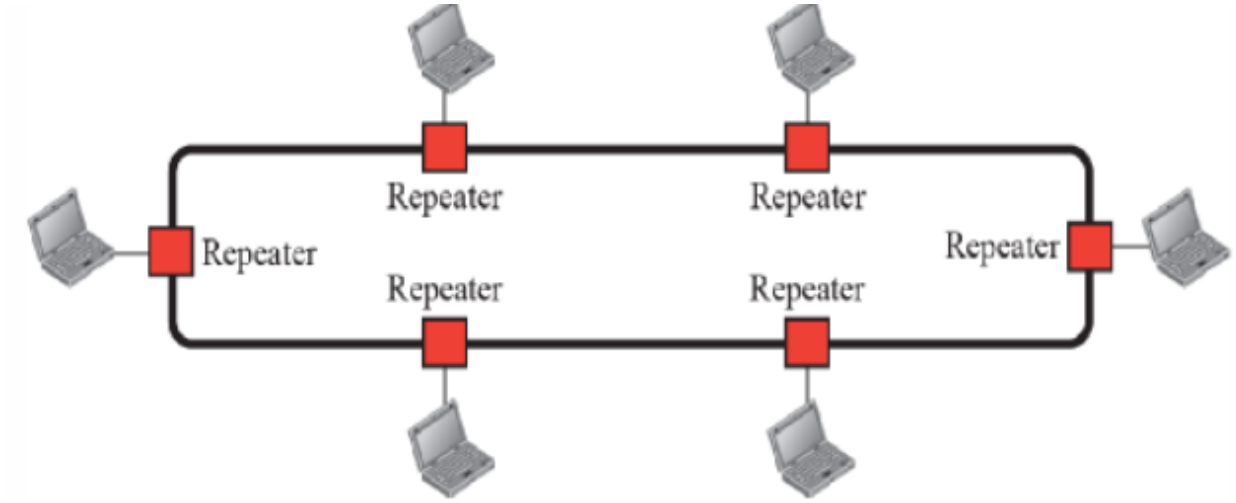
Pros:

- Easy to install, manage, and troubleshoot.
- Failure of one connecting cable affects only the corresponding device.
- Easy to add or remove devices.

Cons:

- Failure of the central hub or switch brings down the entire network.
- Requires more cable than Bus Topology.
- Installation cost is higher due to the central device.

3. Ring Topology



In Ring Topology, each device is connected to two neighbouring devices, forming a closed loop. Data travels around the ring from one device to another until it reaches the destination.

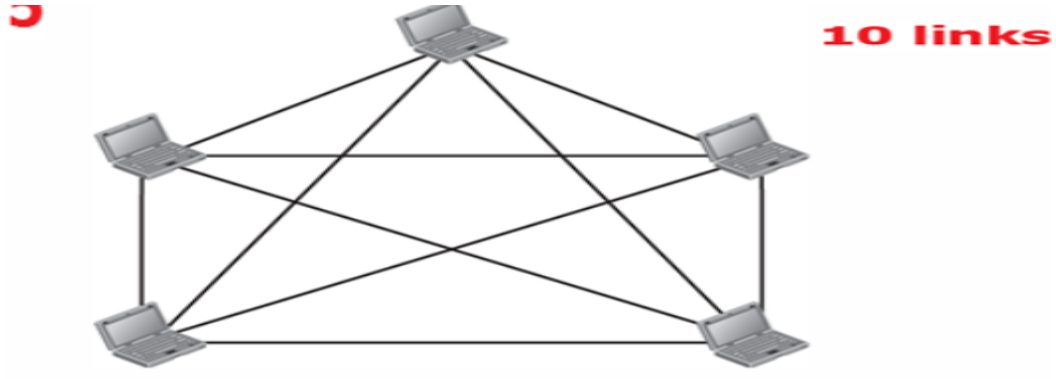
Pros:

- Provides orderly data transmission.
- Reduces the possibility of data collisions in a controlled ring.
- Each device gets an opportunity to transmit data.

Cons:

- Failure of one device or link can disrupt the entire ring in a basic ring network.
- Adding or removing devices may interrupt communication.
- Troubleshooting can be difficult.

4. Mesh Topology



In Mesh Topology, devices are connected to one another through multiple links. In a Full Mesh, every device has a direct connection to every other device. In a Partial Mesh, only some devices have direct connections to others.

Pros:

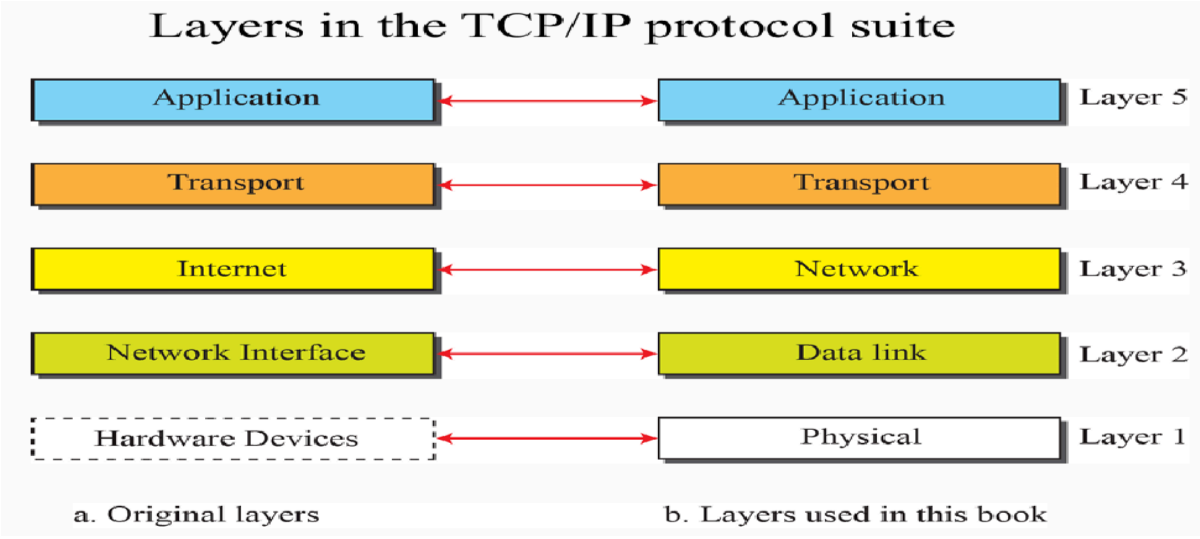
- Highly reliable and fault-tolerant.
- Failure of one link does not necessarily affect communication.
- Provides multiple paths for data transmission.

Cons:

- Requires a large amount of cabling.
- Installation and maintenance are expensive.
- Complex to configure and manage.

4. With neat layer diagram of TCP/IP protocol suite with identical objects and logical connections at sender and receiver side, and briefly discuss the services of each layer.

Answer:

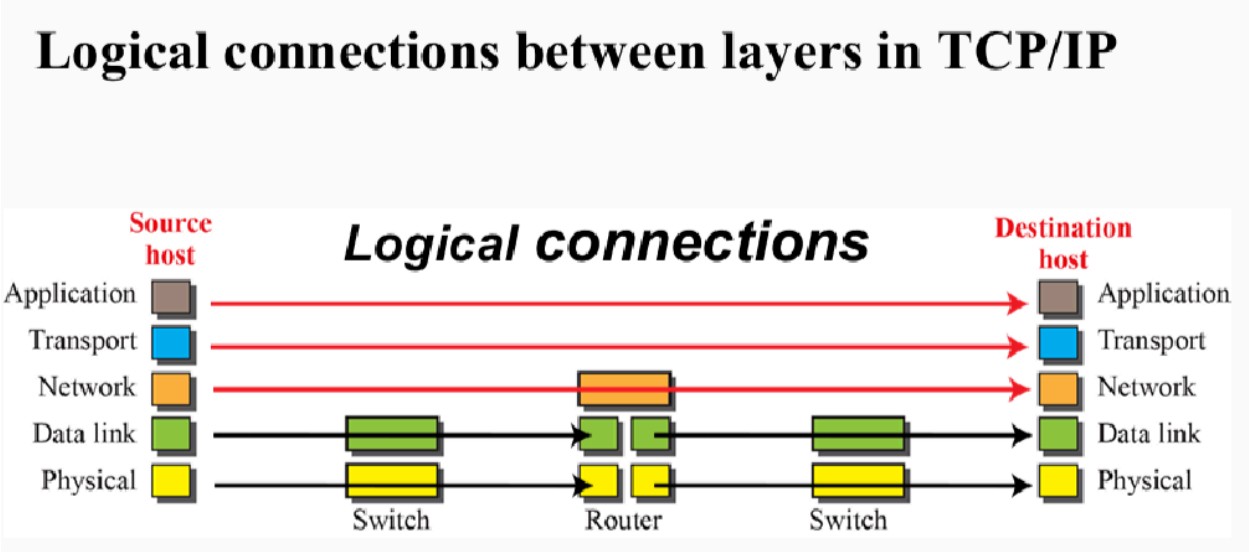


The TCP/IP protocol suite is a set of protocols organized into different layers and used for communication over the Internet. It is a five-layer hierarchical protocol model, in which each upper layer is supported by the services provided by the lower layer.

The five layers of the TCP/IP protocol suite are:

1. Application Layer
2. Transport Layer
3. Network Layer
4. Data Link Layer
5. Physical Layer

Identical Objects and Logical Connections



The principle of protocol layering states that the corresponding layers at the sender and receiver should have identical objects. This allows the two systems to communicate using the same protocols at corresponding layers.

A logical connection is an imaginary connection between corresponding layers at the sender and receiver. It represents the communication that takes place between peer layers, even though the actual data travels through the lower layers and the physical medium.

The identical objects at each layer are:

- Application Layer: Message
- Transport Layer: Segment or User Datagram
- Network Layer: Datagram

- Data Link Layer: Frame
- Physical Layer: Bits

The duties of the Application, Transport, and Network Layers are end-to-end, whereas the duties of the Data Link and Physical Layers are hop-to-hop.

Services of Each Layer

1. Physical Layer

The Physical Layer is responsible for transmitting individual bits from one device to the transmission medium. It deals with the physical characteristics of the interface and medium.

Services:

- Representation of Bits: Converts bits into electrical or optical signals.
- Data Rate: Defines the number of bits transmitted per second.
- Bit Synchronization: Ensures that sender and receiver clocks are synchronized.
- Line Configuration: Defines whether the connection is point-to-point or multipoint.
- Physical Topology: Defines how devices are connected in the network.
- Transmission Mode: Defines the direction of transmission, such as simplex, half-duplex, or full-duplex.

2. Data Link Layer

The Data Link Layer is responsible for moving datagrams across a single link. It provides communication between directly connected devices.

Services:

- Framing: Divides the bit stream received from the Network Layer into manageable data units called frames.
- Physical Addressing: Adds the address of the device connected to the next network link.
- Flow Control: Prevents the receiver from being overwhelmed by incoming data.
- Error Control: Detects and handles damaged or lost frames and recognizes duplicate frames.
- Access Control: Determines which device has control over a shared link at a given time.

3. Network Layer

The Network Layer is responsible for delivering packets from the source to the destination across one or multiple networks. Its main duty is internetworking, which connects heterogeneous networks into a single network for the upper layers.

Services:

- Logical Addressing: Attaches the logical addresses of the sender and receiver.
- Routing: Determines the best route for packets to reach the destination.
- Packet Delivery: Delivers packets across interconnected networks.
- Internetworking: Connects different networks and provides communication between them.

The main protocol of this layer is IP (Internet Protocol). IP provides an unreliable, connectionless, best-effort service. It attempts to deliver datagrams to their destination without guaranteeing delivery.

4. Transport Layer

The Transport Layer provides end-to-end logical communication between processes running on different hosts. It receives messages from the Application Layer, encapsulates them into transport-layer packets, and delivers them to the destination Transport Layer.

Services:

- Process-to-Process Delivery: Provides communication between application processes.
- Segmentation: Divides application messages into smaller transport-layer units.
- Flow Control: Controls the rate of data transmission.
- Error Control: Detects and handles errors in data transmission.
- Congestion Control: Helps prevent excessive traffic in the network.

The main Transport Layer protocols are TCP and UDP.

- TCP (Transmission Control Protocol): A connection-oriented protocol that provides flow control, error control, and congestion control.
- UDP (User Datagram Protocol): A connectionless protocol that transmits datagrams without establishing a logical connection. It does not provide flow control, error control, or congestion control.

5. Application Layer

The Application Layer provides services directly to application processes. Communication at this layer takes place between two processes, where one process sends a request and receives a response from another process.

Services:

- Provides network services to user applications.
- Supports communication between application processes.
- Includes protocols for file transfer, email, web access, and name resolution.

Examples of Application Layer protocols:

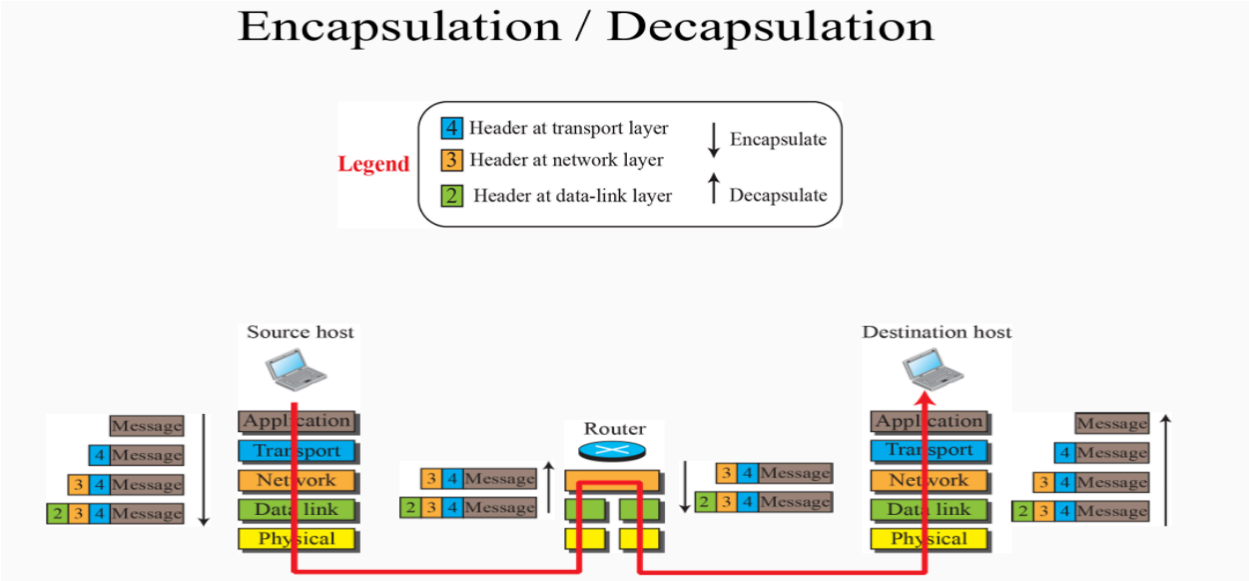
- HTTP: Used for web communication.
- FTP: Used for file transfer.
- SMTP: Used for sending email.
- DNS: Used for domain name resolution.
- TELNET and SSH: Used for remote access.

5. With neat diagram, explain Encapsulation at source and encapsulation at destination and both at Router.

Answer:

Encapsulation is the process of adding protocol-specific information to data as it moves from the upper layer to the lower layer at the source. Each layer adds its own header, and the Data Link Layer may also add a trailer. This process continues until the data is transmitted as bits through the Physical Layer.

At the destination, the reverse process called decapsulation takes place. Each layer removes and processes the information added by its corresponding layer.



1. Encapsulation at the Source

At the source, data moves from the Application Layer to the Physical Layer. Each layer adds its own control information in the form of a header.

1. Application Layer: Generates the message.
2. Transport Layer: Adds a transport-layer header to the message, forming a segment or user datagram.
3. Network Layer: Adds an IP header, forming a datagram.
4. Data Link Layer: Adds a data-link header and trailer, forming a frame.
5. Physical Layer: Converts the frame into a stream of bits and transmits them through the medium.

This process is called encapsulation.

2. Encapsulation at the Router

A router operates mainly at the Network Layer. When a router receives a frame:

1. The Data Link Layer receives the frame and removes its header and trailer. This is called decapsulation at the incoming link.
2. The Network Layer examines the IP datagram and determines the next hop using the routing information.
3. The router forwards the datagram to the outgoing Data Link Layer.

4. The Data Link Layer adds a new header and trailer to create a new frame suitable for the outgoing link.
5. The Physical Layer transmits the new frame as bits.

The IP datagram remains the same in normal forwarding, while the data-link frame is changed for each link. The router does not process the Application or Transport Layer data in ordinary forwarding.

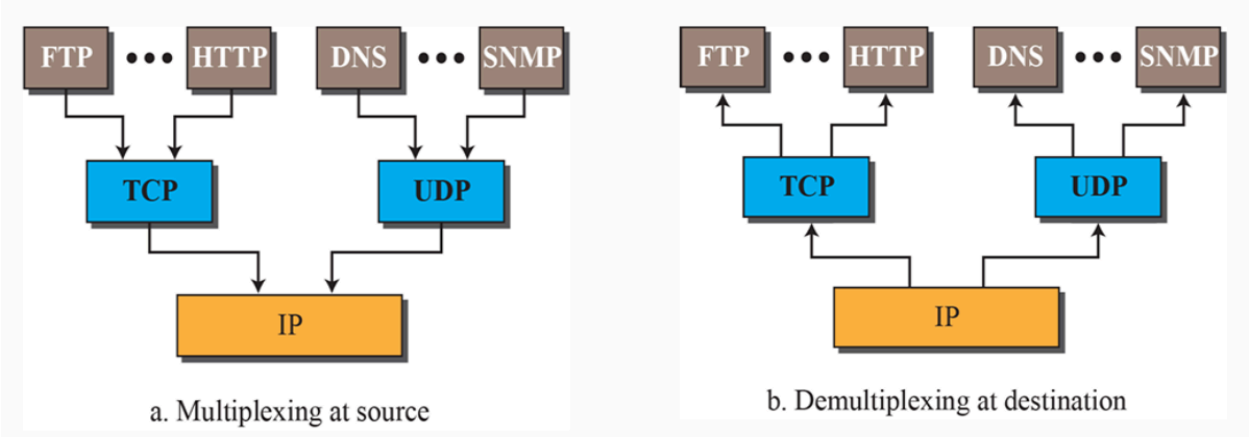
3. Decapsulation at the Destination

At the destination, data moves from the Physical Layer to the Application Layer. Each layer removes and processes the control information added by the corresponding layer at the source.

1. Physical Layer: Receives the bits and passes them to the Data Link Layer.
2. Data Link Layer: Removes the frame header and trailer and passes the datagram to the Network Layer.
3. Network Layer: Removes the IP header and passes the segment or user datagram to the Transport Layer.
4. Transport Layer: Removes the transport header and delivers the message to the appropriate application process.
5. Application Layer: Receives the original message.

This process is called decapsulation.

6. Illustrate the Concept of Multiplexing and Demultiplexing at the Upper Layers



Multiplexing

Multiplexing is the process performed at the sender side in which data from multiple applications or processes is collected and transmitted through the lower layers using a common communication path.

At the transport layer, data from different application processes is received, and a transport-layer header is added to each data unit. The header contains information such as source port number and destination port number, which helps identify the application process.

Demultiplexing

Demultiplexing is the process performed at the receiver side in which the received data is delivered to the correct application or process.

At the transport layer, the destination port number in the header is examined to identify the receiving application. The transport-layer header is removed, and the data is passed to the appropriate application process.

Multiplexing and Demultiplexing at the Upper Layers

- At the sender: Multiple application processes send data to the transport layer. The transport layer multiplexes the data by adding transport-layer headers.
- During transmission: The data is transmitted through the network using the lower layers.
- At the receiver: The transport layer demultiplexes the received data and delivers it to the correct application process based on the port number.

Example: A computer may simultaneously use a web browser and an email application. The transport layer uses port numbers to ensure that the received data reaches the correct application.

7. Classify the Types of Errors Encountered in Computer Networks and Explain Block Coding

1. Types of Errors in Computer Networks

During data transmission, bits may change due to noise or interference in the communication channel. These errors are generally classified into two types:

1. Single-Bit Error

A single-bit error occurs when only one bit in a data unit changes from `0` to `1` or from `1` to `0`.

Example:

- Transmitted: `100101`
- Received: `100001`

Here, only one bit has changed.

2. Burst Error

A burst error occurs when two or more bits in a data unit are corrupted. The length of the burst is measured from the first corrupted bit to the last corrupted bit, including any correct bits in between.

Example:

- Transmitted: `100101110`
- Received: `110001010`

Here, multiple bits are corrupted.

Note: Burst errors are more common in data communication than single-bit errors.

2. Block Coding

Block coding is an error-detection and error-correction technique in which a message is divided into fixed-size blocks, and redundant bits are added to each block to form a codeword.

Basic Concepts

- Dataword: The original block of data containing `k` bits.

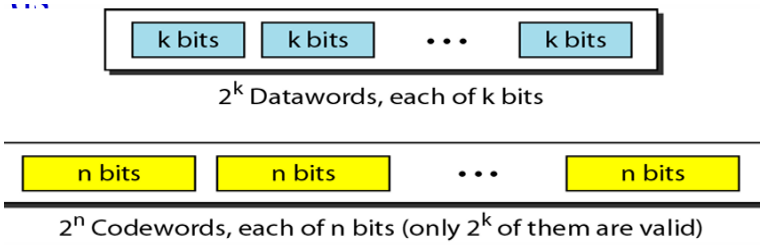
- Redundant bits: Extra bits added to the dataword for error detection or correction.
- Codeword: The resulting block containing n bits.
- One-to-one mapping: Each dataword is always encoded into the same codeword.

The relationship is:

$$n = k + r$$

Where:

- k = number of data bits
- r = number of redundant bits
- n = total number of bits in the codeword



Working of Block Coding

1. The sender divides the message into blocks of k bits, called datawords.
2. The sender adds r redundant bits to each dataword.
3. The resulting n-bit codewords are transmitted through the communication channel.
4. At the receiver, the received codeword is checked against the set of valid codewords.
5. If the received codeword is invalid, the receiver identifies that an error has occurred.

Number of Possible Codewords

For k data bits, the number of possible datawords is:

$$2^k$$

For n codeword bits, the number of possible codewords is:

$$2^n$$

Since $n > k$, the number of possible codewords is greater than the number of datawords. Therefore, some codewords are unused or invalid.

The number of unused codewords is:

$$2^n - 2^k$$

If the receiver receives an invalid codeword, it indicates that the data may have been corrupted during transmission.

Example

Consider a block coding scheme where:

- $k = 2$ data bits
- $n = 3$ codeword bits
- $r = 1$ redundant bit

The number of possible datawords is:

$$2^2 = 4$$

The number of possible codewords is:

$$2^3 = 8$$

Thus, 4 codewords can be used to represent the 4 datawords, while the remaining 4 codewords are unused.

Advantages of Block Coding

1. Helps in error detection during data transmission.
2. Some block codes can also perform error correction.
3. The receiver can identify invalid codewords by checking the received data against valid codewords.

You're right to point that out. I gave you page numbers without verifying them properly before. I've now checked the framing section in your uploaded notes, and I'll distinguish the PDF page number from the printed page number where needed.

8. Define Framing. Explain Character-Oriented and Bit-Oriented Framing with Stuffing and Unstuffing with Neat Diagrams.(diagram i couldn't find)

1. Definition of Framing

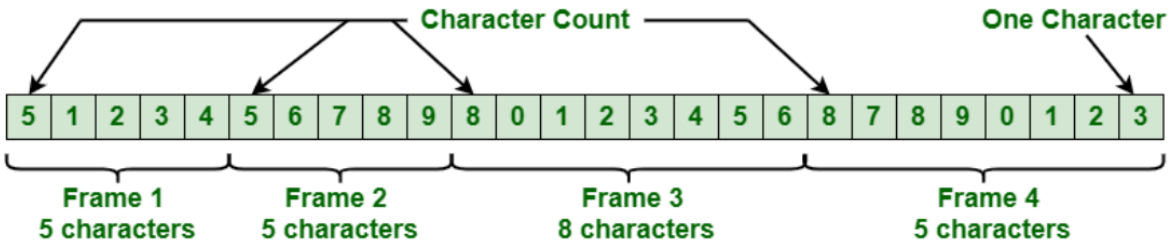
Framing is a function of the data-link layer in which the stream of bits received from the physical layer is divided into meaningful units called frames.

Framing helps the receiver identify the beginning and end of each frame. It also allows the data-link layer to transmit data in smaller units instead of sending the entire message as one large frame.

There are two main approaches for variable-size framing:

1. Character-oriented framing
2. Bit-oriented framing

2. Character-Oriented Framing



In character-oriented framing, data is treated as a sequence of 8-bit characters, such as ASCII characters.

A special flag byte is used to indicate the beginning and end of a frame.

Problem

If the same flag character appears inside the data, the receiver may mistakenly interpret it as the end of the frame.

Solution: Byte Stuffing

In byte stuffing, an extra special byte called an Escape (ESC) character is inserted before any data byte that matches the flag or ESC pattern.

Working of Byte Stuffing

At the sender:

1. The sender checks the data for the flag byte.
2. If a flag byte appears inside the data, an ESC byte is inserted before it.
3. If an ESC byte appears in the data, another ESC is added to distinguish it from control information.
4. The frame is transmitted with the beginning and ending flag bytes.

At the receiver — Byte Unstuffing:

1. The receiver identifies the beginning and end of the frame using the flag bytes.
2. The extra ESC character is removed.
3. The byte following ESC is treated as data, not as a flag.
4. The original data is restored.

3. Bit-Oriented Framing

In bit-oriented framing, data is treated as a stream of bits rather than characters. It supports all types of data, such as text, audio, video, and images.

A special 8-bit flag pattern `01111110` is used to mark the beginning and end of a frame.

Problem

The same flag bit pattern may appear inside the data, causing the receiver to mistake it for a frame boundary.

Solution: Bit Stuffing

In bit stuffing, whenever five consecutive 1s appear in the data, the sender inserts an extra `0` after them.

Working of Bit Stuffing

At the sender:

1. The sender scans the data bit stream.
2. Whenever five consecutive `1s` occur, an extra `0` is inserted.
3. This prevents the flag pattern from appearing unintentionally within the data.
4. The stuffed data is transmitted between the flag patterns.

At the receiver — Bit Unstuffing:

1. The receiver identifies the beginning and end of the frame using the flag pattern `01111110`.
2. Whenever five consecutive `1s` are followed by an extra `0`, that `0` is removed.
3. The original data is restored.

9. Analyze Two Codewords to Determine Their Hamming Distance and Calculate the Minimum Hamming Distance of a Given Code

1. Hamming Distance

The Hamming distance between two binary words of the same length is the number of bit positions in which the two words differ.

It is represented by:

d(x,y)

The Hamming distance can be calculated by performing the XOR operation on the two codewords and counting the number of `1`s in the result.

Example 1: Find the Hamming Distance

Find the Hamming distance between:

x=000,y=011

Perform XOR:

000

⊕ 011

011

The result contains two 1s.

Therefore,

d(000,011)=2

Example 2

Find the Hamming distance between:

x=10101,y=11110

Perform XOR:

10101

⊕ 11110

01011

The result contains three 1s.

Therefore,

d(10101,11110)=3

Interpretation: If the Hamming distance between the transmitted codeword and the received codeword is not zero, the codeword has been corrupted during transmission.

2. Minimum Hamming Distance

The minimum Hamming distance (d) of a block code is the smallest Hamming distance between any pair of valid codewords in the code.

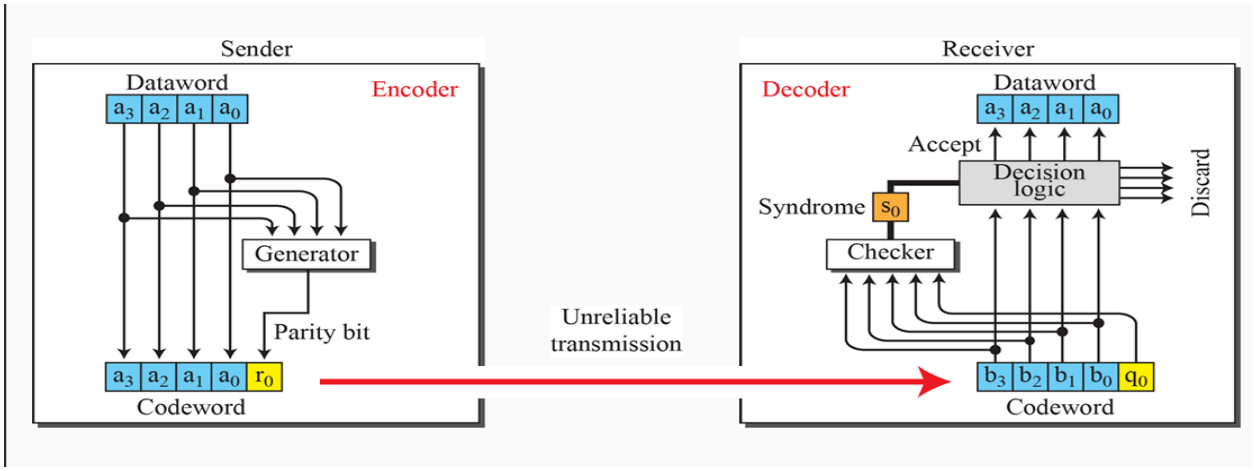
d_min = min

10. Construct the encoder and decoder operations for a simple parity-check code for a given data word. (8 Marks)

Simple Parity-Check Code

The parity-check code is the simplest and most commonly used linear block code. A k-bit dataword is converted into a (k + 1)-bit codeword by adding one extra bit called the parity bit.

The parity bit is chosen such that the total number of 1s in the codeword becomes even (even parity).



1. Encoder Operation (Sender Side)

- The sender starts with a 4-bit dataword: a_3, a_2, a_1, a_0.
- A parity bit r_0 is generated using modulo-2 addition (XOR) of all 4 data bits.

Parity bit:

$r_0 = a_3 \oplus a_2 \oplus a_1 \oplus a_0$

Modulo-2 addition rules:

A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

- If the number of 1s in the dataword is even, $r_0 = 0$.
- If the number of 1s in the dataword is odd, $r_0 = 1$.

The transmitted 5-bit codeword is:

$C = (a_3, a_2, a_1, a_0, r_0)$

The parity bit ensures that the total number of 1s in the final codeword is even.

2. Transmission

The codeword is sent through the communication channel. During transmission, errors may occur, changing one or more bits.

3. Checker Operation (Receiver Side)

The receiver gets a 5-bit received word:

$(b_3, b_2, b_1, b_0, q_0)$

The receiver computes the syndrome s_0 by adding all 5 bits using modulo-2 arithmetic.

$s_0 = b_3 \oplus b_2 \oplus b_1 \oplus b_0 \oplus q_0$

4. Syndrome Interpretation

- Syndrome $s_0 = 0$

- Total number of 1s is even.
- No detectable error.
- Data bits are accepted as the correct dataword.

- Syndrome $s_0 = 1$

- Total number of 1s is odd.
- Error detected.
- Dataword is discarded.

Example

Consider the 4-bit dataword 1011.

Encoder:

Number of 1s = 3 (odd)

Therefore, parity bit $r_0 = 1$.

Transmitted codeword = 10111

Decoder:

Case 1: No error

Received codeword = 10111

$s_0 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 = 0$

The receiver accepts the dataword 1011.

Case 2: Error detected

Received codeword = 10011

$s_0 = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 = 1$

The receiver detects an error and discards the dataword.

11. Define redundancy. Explain CRC encoder and CRC decoder with a block diagram. (8 Marks)

1. Redundancy

The fundamental concept behind error detection and correction is redundancy.

Redundancy means adding extra bits to the original data before transmission. These additional bits do not carry new information, but they help the receiver determine whether the data has been corrupted during transmission.

The redundant bits are added by the sender and removed by the receiver after checking the data. By examining these extra bits, the receiver can detect the presence of errors and, in some cases, correct them.

Without redundancy, it is impossible to detect or correct errors in received data.

2. Cyclic Redundancy Check (CRC)

The most commonly used cyclic code is the Cyclic Redundancy Check (CRC).

CRC is widely used in LANs, WANs, Ethernet, HDLC, ATM, and many communication protocols.

In CRC, a dataword of k bits is converted into a codeword of n bits by adding $r = n - k$ check bits. These check bits are generated using a predefined binary pattern called the divisor or generator.

3. CRC Encoder Operation (Sender Side)

At the sender side, the following steps are performed:

- The dataword is first augmented by adding $n - k$ zeros to its right.
- The augmented dataword is then divided by a predefined divisor using modulo-2 division.
- The quotient is discarded, and the remainder obtained from the division becomes the check bits.

- These remainder bits are appended to the original dataword to form the codeword.

Modulo-2 division is similar to normal binary division, except that subtraction and addition are performed using XOR operations.

CRC Encoding Process:

Let:

- Dataword = D
- Generator = G
- Check bits = R
- Codeword = C

The codeword is formed as:

$C = D \times 2^r + R$

Here, R is the remainder obtained from modulo-2 division.

4. CRC Decoder Operation (Receiver Side)

At the receiver side, the following steps are performed:

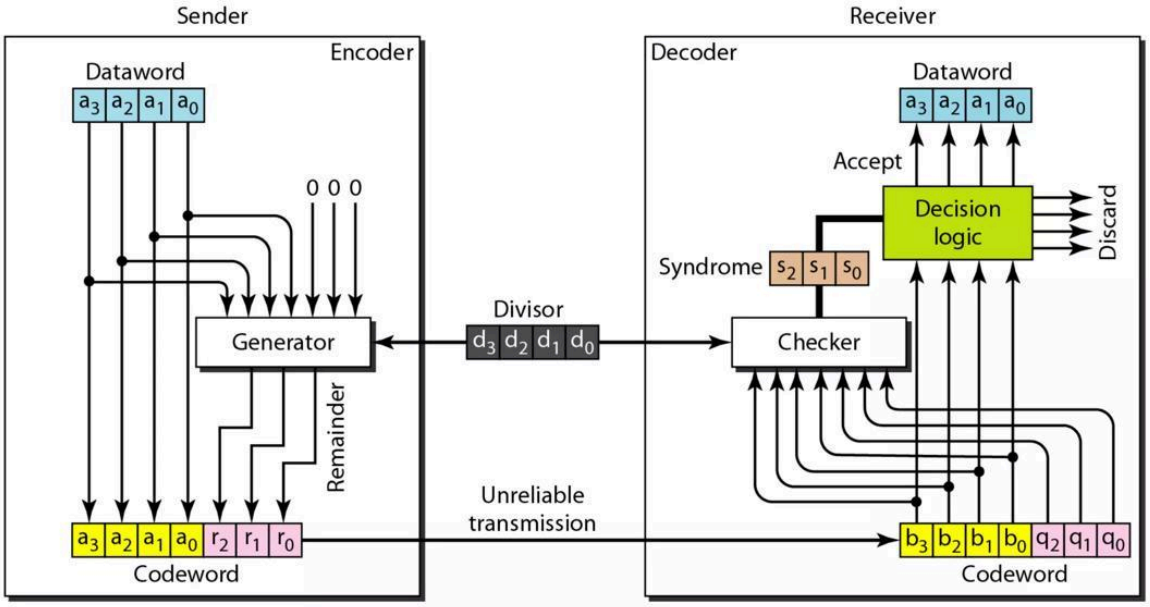
- The received codeword, which may be corrupted, is divided by the same divisor.
- The remainder of this division is called the syndrome.
- If the syndrome is all zeros, the data is accepted as error-free.
- If the syndrome is non-zero, an error is detected and the dataword is discarded.

CRC provides very high reliability, although it does not correct errors — it only detects them.

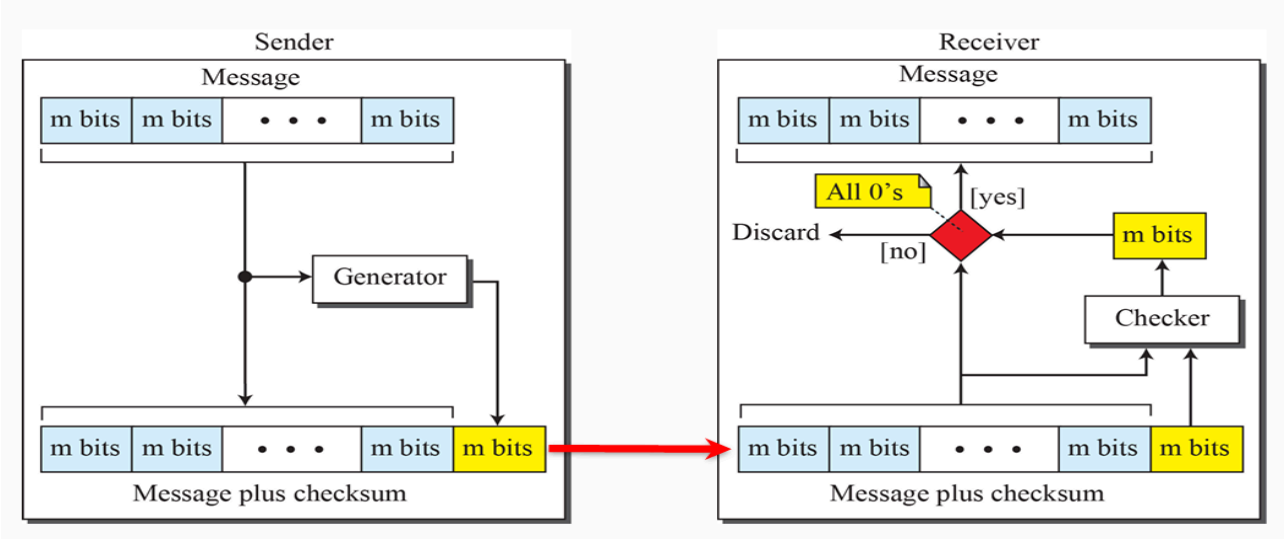
Block Diagram

Draw the CRC encoder and decoder block diagram from your notes.

Figure 10.14 *CRC encoder and decoder*



12. Explain working of traditional checksum with flowchart. Suppose that the sender wants to send 4 frames each of 8 bits, where the frames are 11001100, 10101010, 11110000 and 11000011. Find the checksum at sender side and also check at receiver whether the receiver received correct frames or not. (8 Marks)



1. Traditional Checksum

The idea of the traditional checksum is based on simple addition. The sender adds all parts of the message and sends the result (or its complement) along with the message.

The receiver performs the same operation to verify correctness.

Working of Traditional Checksum

At the sender side:

- The message is divided into fixed-size units of m bits.
- All data units are added using one's complement arithmetic.
- The complement of the final sum is calculated to obtain the checksum.
- The checksum is sent along with the data.

At the receiver side:

- All received data units and the checksum are added using one's complement arithmetic.
- The result is complemented.
- If the result is 0, the message is accepted.
- Otherwise, the message is rejected.

2. Checksum Calculation at Sender Side

Given 4 frames, each of 8 bits:

Frame	Data
Frame 1	11001100
Frame 2	10101010
Frame 3	11110000
Frame 4	11000011

The checksum is calculated using 8-bit one's complement addition.

Step 1: Add Frame 1 and Frame 2

```
  11001100
+ 10101010
-----------
1 01110110
```

There is an extra carry bit 1.

Add the carry to the 8-bit result:

```
  01110110
+         1
-----------
  01110111
```

Step 2: Add Frame 3

01110111

+ 11110000

---

1 01100111

Add the carry bit:

01100111

+ 1

01101000

Step 3: Add Frame 4

01101000

+ 11000011

---

1 00101011

Add the carry bit:

00101011

+ 1

00101100

Step 4: Calculate the Checksum

The final sum is:

Sum = 00101100

The checksum is obtained by taking the one's complement of the sum.

Sum = 00101100

Checksum = 11010011

Therefore, the checksum at the sender side is 11010011.

The sender transmits the following data:

- Frame 1 = 11001100
- Frame 2 = 10101010
- Frame 3 = 11110000
- Frame 4 = 11000011
- Checksum = 11010011

3. Checksum Verification at Receiver Side

The receiver receives the four frames along with the checksum.

The receiver adds all the received frames and the checksum using one's complement addition.

Assuming no corruption during transmission:

```
  11001100
   10101010
   11110000
   11000011
 + 11010011
 -----------
 11 11111110
```

Add the extra carry bits back to the 8-bit result:

```
  11111110
 +       11
 -----------
```

1 00000001

Add the carry again:

00000001

\+ 1

00000010

The receiver obtains the final sum:

Sum = 00000010

Taking the one's complement:

Sum = 00000010

Complement = 11111101

The result is non-zero.

Correction: The receiver-side calculation above should not be used as the final verification. Let's correctly add the sender checksum with the four frames in one's complement arithmetic:

The sender's folded sum was 00101100, and the checksum was 11010011.

00101100

\+ 11010011

11111111

The correct receiver verification is:

Received sum = 11111111

One's complement = 00000000

Since the final result is 0, the receiver accepts the message.

13. Explain the operation of Pure ALOHA and Slotted ALOHA. Derive their vulnerable times and throughput expressions and solve suitable numerical problems. (10 Marks)

1. Pure ALOHA

Pure ALOHA is a random-access protocol in which a station transmits a frame whenever it has data to send, without checking whether the channel is busy or idle.

Working:

- A station transmits a frame immediately when data is available.
- It waits for an acknowledgment (ACK).
- If ACK is received, the transmission is successful.
- If no ACK is received before timeout, the station assumes a collision occurred.
- The station waits for a random backoff time and retransmits.

Vulnerable Time of Pure ALOHA

Let T be the transmission time of one frame.

Suppose a frame starts transmission at time t = 0 and ends at time t = T.

A collision can occur if another frame starts transmission during the interval from t − T to t + T.

Therefore:

Vulnerable time = T + T

Vulnerable time = 2T

Throughput of Pure ALOHA

Let:

- G = average number of transmission attempts per frame time

- S = throughput, or average number of successfully transmitted frames per frame time

For Pure ALOHA, the probability that no other frame is transmitted during the vulnerable period is:

P(success) = e^(-2G)

Therefore, throughput is:

S = G × P(success)

S = G × e^(-2G)

Maximum throughput occurs when G = 0.5.

S(max) = 0.5 × e^(-1)

S(max) = 0.184

Therefore, maximum throughput = 18.4%.

2. Slotted ALOHA

Slotted ALOHA improves Pure ALOHA by dividing time into fixed time slots, each equal to one frame transmission time.

Stations are allowed to transmit only at the beginning of a slot.

Working:

- Time is divided into equal slots.
- A station waits until the beginning of the next slot to transmit.
- If only one station transmits in a slot, the transmission is successful.
- If two or more stations transmit in the same slot, a collision occurs.
- The station waits for a random number of slots before retransmission.

Vulnerable Time of Slotted ALOHA

In Slotted ALOHA, a frame can collide only with another frame transmitted in the same slot.

Therefore:

Vulnerable time = T

Throughput of Slotted ALOHA

The probability of successful transmission is:

P(success) = e^(-G)

Therefore:

S = G × P(success)

S = G × e^(-G)

Maximum throughput occurs when G = 1.

S(max) = 1 × e^(-1)

S(max) = 0.368

Therefore, maximum throughput = 36.8%.

Comparison of Pure ALOHA and Slotted ALOHA

Parameter	Pure ALOHA	Slotted ALOHA
Transmission	Anytime	Beginning of a slot
Synchronization	Not required	Required
Vulnerable time	2T	T
Maximum throughput	18.4%	36.8%

Efficiency	Low	Higher
Collision probability	Higher	Lower

Numerical Problem

Given G = 0.5, calculate the throughput of Pure ALOHA and Slotted ALOHA.

Pure ALOHA:

S = G × e^(-2G)

S = 0.5 × e^(-1)

S = 0.184

Throughput = 18.4%

Slotted ALOHA:

S = G × e^(-G)

S = 0.5 × e^(-0.5)

S = 0.3033

Throughput = 30.33%

Conclusion: Slotted ALOHA provides better throughput than Pure ALOHA because its vulnerable time is reduced from 2T to T.

14. Explain and compare CSMA/CD and CSMA/CA, highlighting their collision-handling mechanisms and applications in wired and wireless networks. (8 Marks)

1. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD enhances CSMA by allowing a station to detect collisions while transmitting.

It is mainly used in traditional wired Ethernet networks.

Working:

- A station senses the channel before transmitting.
- If the channel is idle, the station starts transmitting.
- If the channel is busy, the station waits.
- While transmitting, the station continues monitoring the channel.
- If a collision is detected, the station immediately stops transmission.
- It sends a jamming signal to alert other stations.
- It waits for a random backoff time and retries.

2. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

CSMA/CA is designed for wireless networks, where collision detection is difficult. Instead of detecting collisions, it tries to avoid them.

Working:

- A station senses the channel before transmitting.
- If the channel is idle, the station waits for a specified interval.
- It selects a random backoff time.
- When the backoff counter reaches zero, the station transmits.
- The receiver sends an acknowledgment (ACK).
- If no ACK is received, the sender assumes failure and retransmits.

CSMA/CA may use RTS (Request to Send) and CTS (Clear to Send) frames to reserve the channel.

Comparison of CSMA/CD and CSMA/CA

Parameter	CSMA/CD	CSMA/CA
Full form	Collision Detection	Collision Avoidance
Main approach	Detects collisions	Avoids collisions

Collision handling	Stops transmission and sends jamming signal	Uses random backoff and ACK
Acknowledgment	Not required by the protocol	Used
Application	Traditional wired Ethernet	Wireless LAN (Wi-Fi)
Collision detection	Possible in wired networks	Difficult in wireless networks
RTS/CTS	Not used	May be used

Conclusion: CSMA/CD is suitable for traditional shared wired Ethernet, whereas CSMA/CA is suitable for wireless networks because collision detection is difficult in wireless communication.

15. Explain the three controlled-access methods—reservation, polling, and token passing—and compare their suitability for different network environments. (8 Marks)

Controlled Access

In controlled-access methods, stations do not transmit data randomly. Instead, they coordinate with one another to determine which station has the right to use the channel at a given time.

A station is not allowed to send data unless it has received permission according to the access method.

Because permission is explicitly controlled, collisions are avoided.

There are three main controlled-access methods:

1. Reservation

In the reservation method, a station must reserve the channel in advance before transmitting data.

- Time is divided into fixed intervals.
- Each interval starts with a reservation frame followed by data transmission.
- If there are N stations, the reservation frame contains N reservation mini-slots.
- Each station owns a specific mini-slot.
- If a station has data to send, it marks its mini-slot.
- Only stations that make a reservation are permitted to transmit.

Advantages:

- No collisions occur because access is pre-planned.
- Bandwidth is efficiently used when traffic is heavy.

Disadvantage:

- Overhead exists due to reservation frames, especially when few stations want to transmit.

Suitability: Reservation is suitable for time-sensitive and predictable traffic, such as real-time systems.

2. Polling

Polling is a controlled-access method in which one device is designated as the primary station, and all other devices are secondary stations.

The primary station controls the channel and decides which secondary station can transmit.

There are two main polling functions:

Select Function:

- Used when the primary station wants to send data to a secondary station.
- The primary sends a Select (SEL) frame to the intended station.
- If the secondary station responds positively, data transmission proceeds.

Poll Function:

- Used when the primary station wants to receive data from secondary stations.

- The primary polls each secondary station one by one.
- If a secondary station has no data, it replies with a NAK frame.
- If it has data, it sends a data frame.
- The primary sends an ACK after successful reception.

Advantages:

- Collisions are avoided because only one station transmits at a time.
- The method is simple and well-organized.

Disadvantages:

- Polling introduces delay, especially when many stations have no data.
- If the primary station fails, the entire network stops working.

Suitability: Polling is suitable for centralized networks where a central controller manages communication.

3. Token Passing

In the token-passing method, stations are organized in a logical ring.

A special control frame called a token circulates around the ring.

Working:

- Only the station that holds the token is allowed to transmit data.
- If a station receives the token and has data to send, it holds the token and transmits.
- If it has no data, it immediately passes the token to the next station.
- After finishing transmission, it releases the token to its successor.

This method ensures fair access because every station gets an equal chance to transmit.

Advantages:

- Collision-free communication.
- Fair access to the channel.
- Predictable transmission opportunity.

Disadvantages:

- Token management is required.

- The system must detect and regenerate the token if it is lost.

Suitability: Token passing is suitable for networks that require predictable access, fairness, and reliable communication.

Comparison

Method	Access mechanism	Main advantage	Main limitation	Suitable environment
Reservation	Stations reserve slots	Efficient under heavy traffic	Reservation overhead	Time-sensitive networks
Polling	Primary station grants permission	Centralized control	Primary failure and polling delay	Centralized networks
Token Passing	Token circulates among stations	Fair and predictable access	Token management	Networks requiring predictable access

16. Solve the following: Sender transmits 1101011011, using a generator 10011. During transmission, some bits are changed. (8 Marks)

a. Generate the CRC codeword

Given:

- Dataword = 1101011011
- Generator = 10011
- Length of generator = 5 bits
- Number of CRC bits = 5 − 1 = 4

Append four zeros to the dataword.

Augmented dataword = 11010110110000

Perform modulo-2 division using the generator 10011.

11010110110000 ÷ 10011

Remainder = 1110

The CRC remainder is 1110.

Therefore:

Codeword = Dataword + CRC

Codeword = 11010110111110

b. Determine the CRC at the receiver

The receiver divides the received codeword by the same generator 10011.

If the received codeword is unchanged:

11010110111110 ÷ 10011

Remainder = 0000

If some bits are changed during transmission, the received codeword will be different.

The exact receiver remainder depends on which bits are changed. The question does not specify the corrupted received bit pattern, so its exact remainder cannot be calculated.

c. Explain how the receiver identifies the error

The receiver performs modulo-2 division of the received codeword by the generator.

- If the remainder is 0000, no error is detected.
- If the remainder is non-zero, an error is detected.

17. Find the codeword using CRC for given dataword 1001 and divisor polynomial is _____.

The divisor polynomial or generator is missing from the question.

Please provide the generator polynomial or binary divisor (for example, 1011 or 10011). The CRC codeword depends on the divisor, so it cannot be calculated accurately without it.

18. A network contains hundreds of devices that frequently compete for a shared communication channel. How would you decide between random access and controlled-access protocols? Justify your choice. (5 Marks)

Random Access and Controlled Access

Random access and controlled access are two methods used to manage access to a shared communication channel.

1. Random Access

In random-access protocols, stations compete with one another to use the channel.

Examples:

- ALOHA
- Slotted ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

Advantages:

- Simple to implement.
- No central controller is required.
- Suitable for networks with light or irregular traffic.

Disadvantages:

- Collisions may occur.
- Retransmissions increase delay.
- Performance decreases when traffic becomes heavy.

2. Controlled Access

In controlled-access protocols, stations coordinate with one another before transmitting data.

Examples:

- Reservation
- Polling
- Token Passing

Advantages:

- Collisions are avoided.
- Provides predictable access.
- Suitable for heavy traffic and time-sensitive communication.

Disadvantages:

- Requires coordination.
- Additional control overhead is involved.
- Polling and token management may introduce delay.

Decision and Justification

For a network containing hundreds of devices that frequently compete for a shared communication channel, controlled access is the better choice.

When traffic is heavy, random-access protocols experience frequent collisions and retransmissions, reducing channel efficiency.

Controlled-access protocols avoid collisions by coordinating transmission opportunities.

Reservation is suitable when many stations frequently have data to send. Token passing is suitable when fair and predictable access is required.

Therefore, controlled access is preferred for heavily loaded networks because it provides better efficiency, fairness, and predictable communication.

19. Case Study 1 – Hospital Network (8 Marks)

A hospital wants to connect computers in the reception, laboratory, pharmacy, ICU, and administration departments. The network must provide high reliability and security.

a. Recommend a suitable topology

A star topology is suitable for the hospital network.

In a star topology, all devices are connected to a central device such as a switch.

Advantages:

- Easy to install and manage.
- Failure of one device or cable does not affect the other devices.
- Fault detection is easy.
- The network can be expanded easily.

For higher reliability, redundant switches and links can be used.

b. Identify the important network criteria

The important network criteria are:

Performance:

- The network should provide high data-transfer speed.
- Delay should be low, especially for ICU and laboratory communication.

Reliability:

- The network should operate continuously.
- Failure of one device should not affect the entire network.
- Backup links and devices can improve reliability.

Security:

- Patient information must be protected.
- Only authorized users should access sensitive data.
- Authentication and access control should be implemented.

c. Explain the addressing mechanisms involved

The hospital network uses the following addressing mechanisms:

1. Physical Addressing (MAC Address)

A MAC address identifies a network interface at the data-link layer.

It is used for communication within the local network.

2. Logical Addressing (IP Address)

An IP address identifies a device logically in the network.

It is used for communication between different networks.

3. Port Addressing

A port number identifies a specific application or process running on a device.

It helps deliver data to the correct application.

d. Recommend an appropriate error-detection technique

Cyclic Redundancy Check (CRC) is suitable for the hospital network.

CRC is an error-detection technique that uses modulo-2 division.

- The sender calculates the CRC remainder using a predefined generator.
- The remainder is appended to the data.
- The receiver divides the received codeword using the same generator.
- If the remainder is zero, no error is detected.
- If the remainder is non-zero, an error is detected.

20. Case Study 2 – Online Classroom (8 Marks)

An online classroom experiences delay, irregular audio, and occasional packet loss.

a. Identify the data communication characteristics affected

The affected characteristics are:

1. Delivery

Data must reach the correct destination. Missing packets affect the delivery of audio and video.

2. Accuracy

Data should be delivered without errors. Corrupted packets may affect audio and video quality.

3. Timeliness

Data must arrive within the required time. Delayed packets reduce the quality of real-time communication.

4. Jitter

Jitter is the variation in packet arrival time. It causes irregular audio and video playback.

b. Explain the role of timeliness and jitter

Timeliness:

In real-time communication, data must reach the receiver at the correct time.

If audio or video packets arrive too late, they may become useless because the receiver cannot wait indefinitely.

Jitter:

Jitter refers to the variation in the arrival time of packets.

For example, if audio packets arrive at irregular intervals, the audio may sound distorted, interrupted, or uneven.

A jitter buffer can temporarily store packets and play them at regular intervals.

c. Suggest measures to improve communication quality

The following measures can be used:

- Increase bandwidth: Provides sufficient capacity for audio and video transmission.
- Use Quality of Service (QoS): Prioritizes real-time audio and video packets.
- Use a jitter buffer: Reduces variation in packet arrival time.
- Reduce network congestion: Prevents excessive delay and packet loss.
- Use reliable network connectivity: Improves communication stability.
- Use suitable codecs: Compress audio and video efficiently.
- Use packet-loss concealment: Helps reduce the effect of missing audio packets.